

Wearable computers — an application of BT's mobile video system for the construction industry

T Miah, C Carter, A Thorpe, A Baldwin and S Ashby

The construction industry is acknowledged to be heavily dependent on information flow and communications. Electronic communication from one office to another is relatively straightforward with today's technology. However, when one of the parties wishing to communicate is based out on the construction site, problems of remoteness and insufficient infrastructure can limit the effectiveness of their communications. The emergence of wearable computers offers a potential method of improving this situation, particularly when a visual link is provided by a mobile video transmission system.

This paper outlines the subject of wearable computers, and describes the use of a mobile video camera in a construction application setting. The system has been used on a live construction project at Brindleyplace in Birmingham. Brindleyplace is currently one of the largest commercial construction projects in Europe.

The system is described in detail, followed by some critical comments made by the users of the system. The paper also draws on issues such as possible applications and potential benefits to be gained in the long term. Technology and design issues of the system are also discussed.

1. Introduction

Effective communications are vital in construction due to the large number of project participants, the separation of design and construction disciplines and the geographically dispersed nature of the projects [1]. Numerous studies over the years have highlighted the problems inherent in construction communications [2, 3]. On large construction sites a myriad of interactions between participants take place. These participants include clients, architects, engineers, and a range of specialist contractors, all undertaking construction work, and often contributing to the design as well. The co-ordination of these activities is imperative to the successful completion of a project.

This includes both long-term plans and co-ordination of activities and the short-term day-to-day management of operations on a site. On any single day, numerous queries can arise from the simple delivery of raw materials to major set-backs in the design and layout of parts of the building. These need to be resolved quickly and efficiently if significant overrun costs are to be avoided.

To solve these site queries, personnel on site must collaborate with people off site, namely designers, architects, clients, suppliers and contractors. The solution may not lie with just one person, but with a group of people. Any solution formulated by one person must be checked with others, so that the solution fits the general requirements

of all parties and does not have any knock-on effects on other parts of the design. Even before a solution to a problem can be formulated, the problem must be described to others. Describing the problem to people off site can be difficult, and may involve the use of visual material in the form of diagrams and pictures.

Figure 1 shows the interactions that may occur between site and other participants when queries on site arise. However, the interaction between groups of people around the construction site is also likely to take place as a consequence of queries arising on site (this is not shown in Fig 1).

For example, if the contractor's site team has a query relating to the position or the section of a steel beam, they may first consult the specialist contractor in charge of the steel erection. This may result in one or even both of the parties discussing the arrangement with the relevant design team. The designer may in turn contact the architect if the aesthetics of the building are likely to be affected.

These communications generally require the site staff to repeatedly return to the site office for access to a telephone, or a facsimile for sending sketches of the query. Site visits by the design team often result, and these can sometimes be unnecessary, particularly when a problem is relatively straightforward to solve, but very difficult to describe

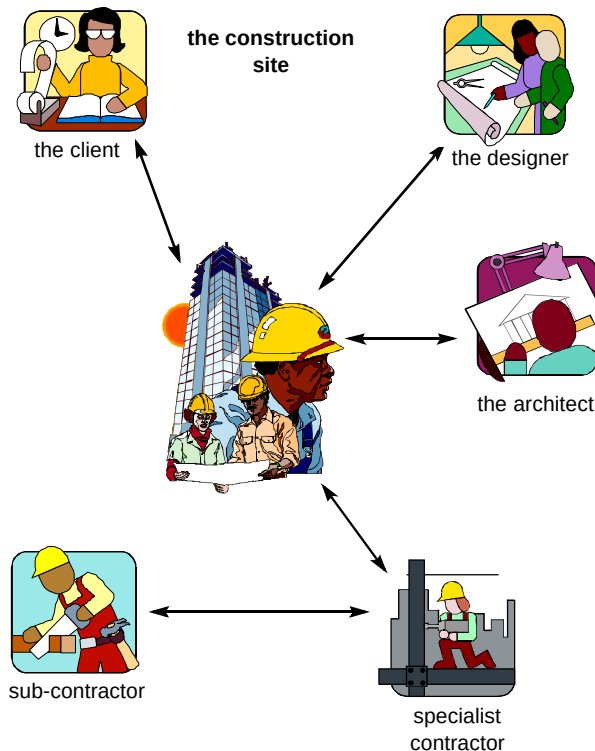


Fig 1 Interaction between major players.

clearly. Modern telecommunications can provide a solution to this, and significantly improve the quality of communications from on site to the design team. Wearable computers can potentially provide this link, involving the use of live video images from the construction site, shared data and access to drawings.

This paper discusses the background to wearable computers and their development. One particular aspect of wearable computers, mobile video transmission, is of particular interest. The paper also describes the work on mobile video and its application to the construction industry currently being conducted on an IMI (Innovative Manufacturing Initiative) project called ISOCCrates (Improving Supply Chain Communication in Construction). The project aims to examine the benefits that can be gained through improved communication between construction supply chain participants.

Technology for sending live video images from the site is described, including the system installed on the Brindleyplace project. The use of this mobile video system on the project is illustrated, with comments from the users, along with other potential applications. Recommendations for further development of the video system are made, and developments in the field of wearable computing related to construction communications are recorded.

2. Wearable computers

The convergence of new and affordable communication and computing technologies is enabling ever more innovative forms and uses of these technologies. One such

development is the emergence of 'wearable computers' which comprise a mobile video camera, a small PC, head-mounted display, and some form of alternative input device to that of a keyboard. Wearable computing goes beyond the traditional mobile or desktop computing concept. By combining ideas from mobile and other computing technologies with advanced user interfaces, such as gestures or voice recognition, wearable computing promises to radically change the way we use computers. Fickas et al [4] outline the following four elements as fundamental to wearable computers.

- Hands-free or one-handed operation

A major advantage of using a wearable computer is not having to use both hands to operate it, and being able to use it while doing other things like holding a tool. Hands-free operation represents an important user-interface design challenge, which has not yet been adequately addressed. Two major sectors that are likely to benefit from the use of wearable computers are the manufacturing industry and the construction industry. For operators from both of these industries, hands-free operation is crucial.

- Mobility

Wearable computers are intended to be used in the real world while doing real-world tasks. Mobility is defined as the user's ability to move around freely while having uninterrupted access to the services of the wearable devices.

- Augmented reality

An augmented-reality system uses a heads-up display to project information on to physical objects in the user's surrounding. This is one component of what is known as mixed reality (MR), the other being augmented virtuality (AV). Anumba et al [5] suggest the emergence of augmented reality, virtuality and other advanced visualisation techniques will improve communication between the designers and their tools. These technologies will enable designers to better understand potential problems, avoiding them before major cost is incurred. The CICC ('kick') project (Collaborative Integrated Communication for Construction) is also looking at innovative uses of the Internet, multimedia and augmented reality [6].

- Perception

Perception refers to the computer's ability to perceive and act upon certain aspects of its physical environment. Perception is the basis for context-dependent computing where the result of computations depends on an explicit set of context variables. The context might include physical and logical parameters such as absolute location, relative location (proximity), identity of objects and persons in the immediate

surrounding, a user's medical condition (heart rate, blood pressure), and the state of devices of machines (temperature, noise level).

Wearable computer systems with telecommunications facilities can allow field workers to access information and contact remote experts. This can be valuable in many settings, from airline or vehicle maintenance, to rural/home health care, to on-the-job training for telephone repair people [7], or resolving construction site queries. The BRICC (Broadband Integrated Communications for Construction) project illustrated this concept by the trial application of mobile video for construction communication [8].

3. The technology

The technology required for mobile video is available, although it is not currently integrated into a single package. The scenario for this requirement of the mobile video system is centred on the construction site. The aim is for construction workers to be able to walk around a construction site with a mobile video camera transmitting live video to a remote participant (see Fig 2). The remote participant may be any other party involved on the project — senior managers, designers, experts in the field, or the client. All of these people have an ongoing requirement to visit and view the site and would therefore gain from remotely seeing the site live as work progresses. Site personnel may have queries on particular aspects of a building and may wish to discuss the design with the architect concerned, also utilising the live video.

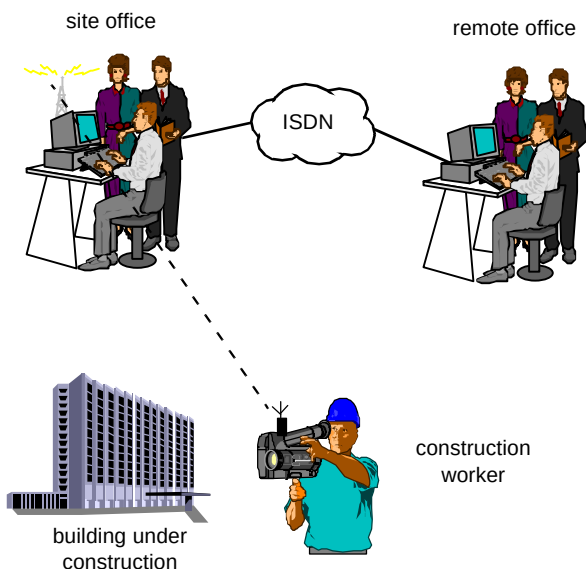


Fig 2 The mobile video system scenario.

The basic components of the system are as follows:

- camcorder/camera with a transmitter, including audio.

- receiver at the site hut that receives the signal and transmits it via a videoconferencing (VC) system anywhere in the world.

3.1 Key requirements for the mobile system

The system requires the following:

- licence-free operation — the system should not require that a radio licence be obtained before operating, as these add cost and inconvenience to the set-up process (licences are also site based, i.e. location specific, restricting free movement to other construction sites),
- good quality — video and audio should be concomitant with the quality of the videoconferencing link (H.320, 64 to 384 kbit/s),
- low cost — the system should not add an excessive cost penalty to a typical installation (PC + videoconferencing kit + ISDN2),
- plug and play — the system should be compatible with all major videoconferencing units,
- maximum portability — the system should be small, light, and show good battery life (similar to a modern camcorder),
- maximum flexibility — the system should accommodate a wide range of applications, with an adequate range.

3.2 The analogue solution

This solution uses an analogue wireless video transmitter to relay the video signal to the site hut. The analogue signal arrives at the receiver and is fed into the videoconferencing unit, which makes it available for transmission using a videoconferencing system (see Fig 3).

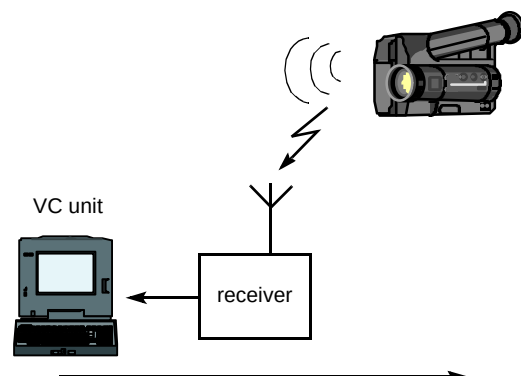


Fig 3 The analogue mobile video system.

Fig 3 The analogue mobile video system.

The system shown in Fig 4 is currently employed at Brindleyplace and does require a radio transmission licence.



Fig 4 The kit used at Brindleyplace.

However, a digital option is also available and a prototype system is currently being developed.

3.3 The digital solution

Rather than using dedicated wireless video transmitters, BT's RealView system (see Fig 5) currently uses the wireless local area network (LAN) technology, which is the European standard for wireless local area networks operating in the industrial, scientific and medical band (2.4 GHz). This provides the potential for low-cost, licence-exempt, wireless delivery of services requiring bit rates of up to 1 Mbit/s.

The mobile part of the RealView system consists of a standard laptop PC, with an adapter box for an M-JPEG (motion JPEG) video card. The video is encoded by the M-JPEG card, and software written by BT sends the packets of encoded video image across the wireless network to the remote LAN connection (in the site office). The wireless LAN card in the mobile unit is a PCMCIA type II card, which is a standard card used for wireless data networks.

The remote LAN end receives these packets and uses an M-JPEG card to decode the packets, returning them to an analogue video signal. The analogue video signal is then fed into the videoconferencing unit, which then operates as a conventional conferencing unit, i.e. one videoconferencing unit to another over an ISDN network.

Figure 6 shows a picture of the actual system. The size of the system is dependent on the size of the notebook. It is still relatively large but, as notebook size reduces, the system should become smaller. This prototype system uses an adapter box connected to the laptop to plug in a full-size M-JPEG card, hence the increase in size. However, as PCMCIA M-JPEG cards become available the system should reduce in size. Also, as the processing speeds of palmtops and PDAs increase it should be possible to use these devices for remote video transmission.

4. The system

The prototype analogue mobile video system described in section 3.2 has been installed on the Brindleyplace site for testing and evaluation of potential applications. It

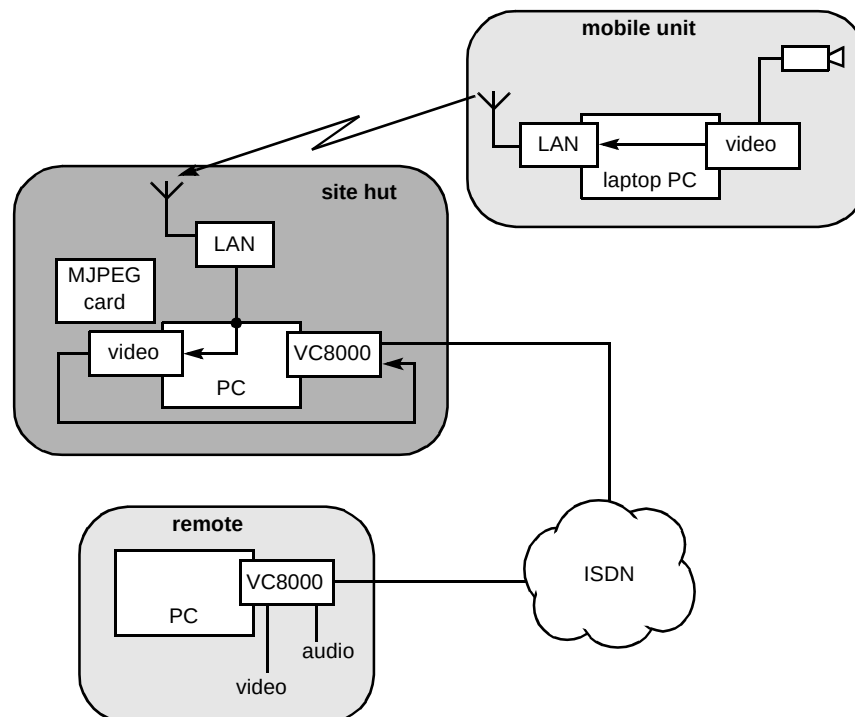


Fig 5 BT's RealView system set-up.



Fig 6 Prototype digital mobile video system.

provides real-time video access to site from remote locations, using a videoconferencing link. Some of the trials on the pilot study and other potential applications of this technology, which can be driven from either location, are discussed below.

4.1 Applications

There are many applications for the mobile video system, both in construction and other industries. The motor insurance industry is already using similar technology to evaluate damage to cars before approving repairs. Within construction, examples of applications include the following.

- Discussion of site queries

The site team can effectively illustrate a query they have for the design team using video images. This will not replace a fax with the relevant dimensions, but does allow clear visualisation of a problem by the design team. One site engineer stated: “.. the link will reassure me that the design team are solving the actual problem I have, not the problem they think I have.”

- Requests for information from the design team

The design team can request footage of specific areas of site in preparation for undertaking further design work or amendments.

- Progress reviews

The equipment can be used on a regular basis to convey overall site progress to the design team. Not all members of the design team have regular access to the site, and previous research highlighted this as a valuable application [9]. A regular progress review is

planned for the next project on the Brindleyplace development, subject to an improvement in the audio link to the unit.

- Record of events

The images from the mobile video system displayed at the remote location can be captured and stored as graphics files. These provide a record of events, which can be incorporated into other documents.

Figure 7 shows the equipment being used at Brindleyplace to examine a stanchion base on one of the buildings under construction.



Fig 7 Mobile video in action.

4.2 A case study of actual use

The mobile system was used on site to review and record a complex location, to enable a remote designer to commence drawing work.

The contracts manager, based on site, used the camera to pan and scan the locality of the roller shutters to be installed at the basement car park entrance. At this location there were off-set columns, beams and block walls which were fire protected. An element of secondary steelwork was required to build out the existing steelwork to receive the roller shutter frame. Existing service trunkings and fire alarm cables had already been installed and would have to be removed and rerouted. Furthermore a division screen and other services and trades would have to be co-ordinated around the roller shutters to be installed.

The roller shutter installation also had associated fixtures and fittings to enhance its operation, with remote links to the fire alarm, security desk, and building management services. Such installation routes have to be co-ordinated with the local structure.

The designer, based in a remote design office, was ready to commence the design work for this area but was not due to visit the site for several weeks. The drawing work needed

to be progressed to tie in with the lead-in periods for the design and fabrication of the roller shutter installation. Therefore, the mobile video camera facility was used to transmit the general layout of the area without the designer having to attend site. This helped the designer visualise the area and make notes of certain constraints that needed to be taken into consideration for the successful subsequent design.

5. General user comments

There are effectively two different users of the system — the site user and the remote viewer. The site user is responsible for taking the camera around site and filming objects of interest. The remote viewer can be a designer, an architect or the client, who is interested in some aspect of the work on site.

Each user has a different perspective of the system and comments on the system from their viewpoint are detailed below.

5.1 Site user

The following technologies must be familiar to the site user for operation:

- desktop videoconferencing facility including camera mounted on monitor and ISDN line,
- receiver, cable linked to VC facility,
- standard camcorder,
- transmitter incorporating battery pack, to be connected to camcorder.

The following procedure is carried out by the site user for live transmissions:

- fix transmitter, battery pack to camcorder and connect leads,
- make video link to remote station,
- site videoconferencing station to be set to auxiliary/self-view mode,
- the only audio link with the remote station is currently by means of a separate mobile phone, i.e. the remote VC user must telephone the mobile separately and the camcorder operator carries the phone in his other hand,
- site user to walk around slowly, as sudden movements result in break-up of the signal to the remote videoconferencing unit.

The facility has a good range (up to 200 metres), with minimal loss of signal behind other buildings.

The camcorder needs to be held steadily — provided the holder moves slowly, the remote viewer receives adequate feedback and pictures.

The zoom facility on the camcorder is useful, as this reduces the camcorder user's movement. However, 'zooming in' increases the apparent picture shake. The battery pack that is fixed to the transmitter is currently too cumbersome.

Figure 8 shows the relative size, mounting of the transmitter, and operational position of the camera. The transmitter/battery pack needs to be fixed to a belt or alternative means of weight support.



Fig 8 Relative size and mounting of transmitter.

5.2 Remote viewer

The mobile video camera has great potential for solving real problems over large distances without the need for lengthy travel; unfortunately, as with most modern technologies, people are initially reluctant to use it.

The quality of the images viewed at the remote location are dependent on a number of factors. Pictures received in the site office can be unstable due to rapid movement of the camera. The wireless transmission system's bandwidth also affects these images, although this is minimal. Degradation results from the coding algorithm, but the overriding factor is the transmission over ISDN lines. The number of these

lines used by a videoconferencing unit, in this case two channels (128 kbit/s), dictates overall quality. For this reason, a video transmitted to London will have a lower image quality than the same video viewed locally in Birmingham.

A good use for the system is to take still 'snapshot' images from the video and relay them to a remote viewer. Good quality pictures are achieved when the camera is held still. However, a simple digital camera would provide a more suitable alternative for capturing still images.

During a series of IT awareness presentations, the system was demonstrated to a number of potential users. Most people were very impressed by the technology, but few could see that its possible use justified the costs.

6. Problems, difficulties and limitations

The primary limitation of the current system is that it is too heavy to use comfortably. The packaging of the system needs to be redesigned so that the transmitter is not bolted on to the camcorder. The transmitter can easily be separated and strapped around the waist. This will take some of the weight from the operator's hand, ensuring more comfortable use of the system.

The current system does not have an integrated audio link built into the system. Instead, a mobile phone is used for the audio connection. This leads to two problems. Firstly the cost is increased unnecessarily (ISDN charges + mobile phone charges), as the absence of the audio on the local link requires a separate audio connection to the remote party, instead of utilising the existing ISDN link. Secondly, if the mobile person is carrying the camera in one hand and a mobile phone in the other, it makes access around site difficult. Ideally, the system should be hands free or limited to one-handed operation so the worker has maximum flexibility, in keeping with the guideline suggested by Fickas et al [4].

To improve the picture quality, a tripod or monopod could be used to hold the camera steady. This would greatly improve the quality of the images, as any slight movement when carrying the camera causes the picture to break up.

The greatest benefit can be gained from the mobile system by making it available to all parties on the project, i.e. contractors, specialist contractors, the client, etc. The mobile video camera should be seen as a central resource for a project. However, this accessibility to the equipment can be a problem. It needs to be in a central location, available to anyone who needs it, while also being stored securely.

7. Future developments

The following list gives details of the items identified as being in need of improvement:

- camcorder users need to be able to hear and talk to the remote station (i.e. headphones and mouth piece),
- the weight of the hand-held devices needs to be reduced,
- smaller battery pack — even if it means less transmission time,
- battery pack should be supported separately (belt),
- VC facilities should be variously located so that remote users can gain maximum benefit.

8. Conclusions

The analogue system used for the trial showed the potential of such a system. However, in the long term the system is likely to move towards the digital solution. This provides for many more applications, as the mobile worker now has access to the features and functionality provided by the laptop.

A possible extension to the system is to use a pen PC for the digital video solution. The wireless LAN capability could be utilised for collaborative work between the mobile worker and a colleague based in the site hut, even a remote expert connected via an ISDN network from the site hut (see Fig 9). This would enable the remote expert to access real-time video of site progression and both parties (remote expert and mobile site worker) could jointly review drawings. Both could look at the schematic layout of a building on the PC, as well as viewing the actual building on video.

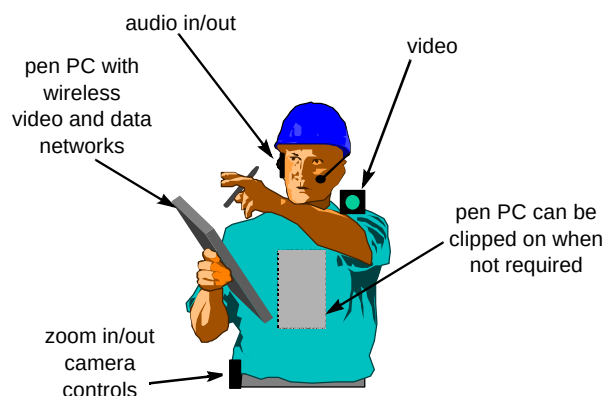


Fig 9 A wearable computer for construction engineers.

The packaging of the system needs to be thought out clearly, taking into account ergonomic issues. It is much

easier for people to carry weight that is strapped around the waist than to hold it in their hands. The system must be hands free, as construction workers on site are likely to climb up ladders, crawl through ducting, etc. The location and fixing of the camera also has to be examined. The multimedia hard hat [10] had a small camera mounted on the front of a safety helmet. The shoulder may be a more suitable position, being less prone to rapid movement and shaking. Another possibility is the wrist, with the video image focused on an object which is being pointed at.

Another interesting aspect to consider is the use of multiple cameras. The views from these would be automatically selected depending on the user's actions. For example, the view from the wrist camera could be selected when the hand was extended horizontally.

Batteries add size and weight, and inconvenience present-day mobile computers. Starner [11] explored the possibility of harnessing some of the energy expended during the user's everyday actions to generate power for their computer. This now appears to be a viable solution.

This paper has investigated the mobile video aspect of wearable computers. A viable solution for some of the construction industry's communications requirements, using currently available mobile video technology, has been successfully demonstrated. Continuing technological advances will lead to the development of a realistic mobile, wearable, computer system for the construction industry.

There is a genuine need for mobile systems for construction workers, and the potential for long-term benefits. However, systems must be ruggedised for use on construction sites and users educated to its full benefits.

References

- 1 Barrie S and Paulson B C: 'Professional construction management: including CM design-construct, and general contracting', 3rd Ed, McGraw-Hill, London (1992).
- 2 Higgin G and Jessop N: 'Communications in the building industry: the report of a pilot study', Tavistock Publications (1965).
- 3 Latham M: 'Constructing the team: joint review of procurement and contractual arrangements in the United Kingdom construction industry', Final report, HMSO (1994).
- 4 Fickas S, Kortuem G and Segall Z: 'Software Issues in Wearable Computing', CHI Workshop on Research Issues in Wearable Computers, Atlanta, GA (March 1997).
- 5 Anumba C J, Baron G and Duke A: 'Information and communications technologies to facilitate concurrent engineering in construction', *BT Technol J*, 15, No 3, pp 199–207 (July 1997).
- 6 Russell L: 'Kicking in: Computing and Communications technology are converging', *New Civil Engineer*, (October 1996).
- 7 Smailagic A and Siewiorek D: 'A case study in embedded system design: VuMan2 wearable computers', *IEEE Design and Test of Computers*, 10 (September 1993).
- 8 Thorpe A, Baldwin C, Carter C, Leevers D and Madigan D: 'Multimedia Communication in Construction', *Proc of the Institution of Civil Engineers*, 108, No 1 (1995).
- 9 Carter C, Thorpe T and Rattee R: 'Multimedia communications in construction: Barking Power Station BRICC pilot phase 2: Implementation issues', Internal reports from the BRICC project (1994).
- 10 Bowen D: 'Builders use their heads with a hi-tech hard hat', *Independent on Sunday* (15 May 1994).
- 11 Starner T: 'Human Powered wearable computing', *IBM Systems Journal*, 35, Nos 3 and 4 (1996).



Tunu Miah received his MEng in Electronic Systems Engineering from Kingston University in December 1992.

He then joined BICC plc, where he worked for the Multimedia Communications research group. He joined LUTCHI Research Centre, Department of Computer Studies, Loughborough University in January 1994, since when he has been working as a research associate on a number of research projects in the field of AI, HCI and CSCW.

He is currently in the process of completing his PhD at Loughborough University.



Chris Carter graduated from Loughborough University in 1989. He was a senior site engineer, before returning to Loughborough to work on the BRICC (Broadband Integrated Communications for Construction) project.

He spent a year at Barking Power Station investigating the application of multimedia communications to the construction industry. A research project examining work process changes resulting from the implementation of IT followed. This included the study of EDI, e-mail, and DIP, along with simulation

techniques for evaluation.

He is currently working on the ISOCCrates (Improving Supply Chain Communications in Construction) project.



Tony Thorpe, BSc, MSc, PhD, CEng, MCI0B, MBCS, is Professor of Construction Information Technology.

His research interests include the applications of advanced IT and communications technologies to construction organisations and the construction process. This has included funded work in the areas of AI, EDI, bar coding applications, multimedia and CSCW. He is immediate past chair of the Association of Researchers in Construction Management (ARCOM).



Andrew Baldwin, BSc, MCs, PhD, CEng, MCIQB, Eur Eng is a Senior Lecturer in Construction Management. His experience in the construction industry includes site engineering, design office practice and planning.

He has also developed construction management software, and worked as a senior systems analyst. His main research interests are work process changes in the construction industry, their impact upon the industry's organisations and the management information systems required to fully benefit

from the 'information revolution'.



Stuart Ashby has been with BT Laboratories since 1989, working mainly with high performance radio communications systems. He has been working on telecommunications and computer applications for low-cost radio technology, particularly multimedia delivery over wireless LANs. Prior to this he has worked on standards development within the European funded RACE program, to develop a Mobile Broadband Service (MBS) specification for future radio delivery of ATM. This work grew from a background in GaAs microwave integrated circuit design, and novel radio/optical component and

systems research. He has now moved on to the BT Wireplay on-line computer games development team.